

Research Graph City Progress Report

Project progress summary and future roadmap, July to September 2026

This report summarizes the Research Graph City prototype built as a sibling app to the original Graph-Cities project. The goal was to transform research-paper graphs into a navigable 3D city where paper clusters become buildings, dense substructure becomes floors, and cross-topic relationships become bridges or streets.

Executive Summary

- Created a new full-stack app in ResearchGraphCity without modifying Graph_City_Web.
- Implemented OpenAlex ingestion, graph construction, building/floor/bridge/street generation, FastAPI endpoints, and a React Three Fiber city renderer.
- Scaled the default research city from a 300-paper first slice to a 1000-paper dataset while keeping the existing inspector, summaries, and rendering behavior working.
- Added seeded-city generation so a user can paste a small list of papers and build a new city from related OpenAlex papers.
- Integrated Groq for structured JSON summaries and navigation answers, with graceful unavailable states when keys or network access are missing.

Current Data

Default Research City

- Seed queries: graph visualization, GraphRAG, and education income mobility.
- Current processed scale: 1000 papers, 6457 paper-level graph edges, 13 buildings, 65 floors, 5 strong bridges, 12 streets, and 8 semantic communities.

Seeded City

- Input: simple textarea where each line is a paper title, DOI, or OpenAlex URL.
- Pipeline: resolve seed papers, expand through related OpenAlex works, dedupe, then run the same graph-city rules as the default city.
- Latest verified seeded run: 991 papers, 7668 graph edges, 12 buildings, 32 floors, 7 bridges, 11 streets, and 8 communities.

Paper Metadata Used

- OpenAlex ID, DOI, title, abstract-derived inverted index, publication year, venue, topics, keywords, authors, institutions, references, citation count, open-access signal, code/data signal, and seed provenance.

Backend Work Completed

- Added FastAPI and Pydantic schemas for vertices, edges, buildings, floors, bridges, streets, communities, summaries, seeded-city requests, and navigation responses.
- Updated OpenAlex access to the current api.openalex.org API style, including API-key

support, per-page limits, seed lookup, deduplication by DOI and OpenAlex ID, and cache-friendly JSON output.

- Built paper-level edges from citations, topic overlap, venue overlap, author/institution overlap, method/dataset overlap, abstract/title text similarity, and temporal proximity.
- Generated buildings from NetworkX graph structure using community detection and core-number signals. Small or isolated groups are preserved as outskirts rather than being lost.
- Generated floors from core-number bands, with a fallback split so buildings with uniform core values still display visible floors.
- Generated bridges for strong cross-building relationships and streets for weaker but still nonzero research relationships. Zero-score layout fallback streets were removed from the inspectable data.
- Exposed APIs for city metadata, buildings, bridges, streets, communities, building details, paper nodes, internal edges, bridge cross edges, seeded-city generation, and LLM summaries/navigation.

Frontend Work Completed

- Built a Vite, React, TypeScript, React Three Fiber, Zustand, and Tailwind frontend.
- Rendered buildings as stacked cylindrical/frustum floors, with labels, grid navigation, zoom controls, camera interaction, bridges, streets, layer toggles, and selected-item highlighting.
- Added building, bridge, street, and paper-node inspectors with metrics, original labels, paper titles, internal edges, cross edges, floor filters, and relationship evidence.
- Added building interior mode so users can inspect paper nodes and edges inside a selected building without crowding the main city view.
- Changed internal and cross-edge displays from opaque R_000001 IDs to paper titles plus metadata, making the evidence readable during demos.
- Added summary-word selection so a word selected in a generated building summary highlights matching labels from the building's original data.
- Added a simple seeded-city panel with a sample button and clearer errors for OpenAlex/network/key failures.

Visual Semantics

- Building: a cluster/community of related papers.
- Footprint: number of papers in the building.
- Height: structural importance from core score plus paper count.
- Floor: a core-number band inside that building.
- Base color family: semantic domain, such as health, networks, economics, biology, or AI.
- Shade variation: separate communities within the same semantic domain get related but distinguishable colors, reducing repeated identical building domains.

- Floor shade: higher-activation floors appear stronger within the building color family.
- Glow: activation score only. It does not change semantic color or building size.
- Bridge: a strong cross-building research relationship.
- Street: a weaker, nonzero relationship that still helps explain city navigation.
- Amber ring and label outline: currently selected building, bridge, street, or paper node.

Metric Definitions

Profile similarity: $0.30 J(\text{topics}) + 0.20 J(\text{methods}) + 0.15 J(\text{datasets}) + 0.10 J(\text{venues}) + 0.10 J(\text{institutions})$, where J is Jaccard similarity.

Semantic similarity: $0.50 J(\text{building labels}) + 0.50 J(\text{profile topics and keywords})$.

Cross-edge strength: $\min(1, \text{cross_building_edge_count} / \sqrt{\text{source_node_count} * \text{target_node_count}})$.

Vertex overlap: $J(\text{vertices_A}, \text{vertices_B})$. This is normally 0 because buildings are disjoint paper sets.

Activation similarity: $1 - \text{abs}(\text{activation_A} - \text{activation_B})$. Higher means the two buildings have similar current activity.

Activation score: $0.55 * \text{recency} + 0.45 * \text{citation_signal}$, where $\text{recency} = \exp(-\text{age}/5)$ and $\text{citation_signal} = \min(1, \text{average_citations} / 100)$.

LLM confidence: A structured-output confidence field returned by Groq for summaries or navigation. It is treated as a model self-assessment and should later be calibrated against retrieval evidence.

LLM Integration

- Groq is called through its OpenAI-compatible API and asked to return structured JSON, not free-form text that the app has to guess how to parse.
- The LLM receives compact evidence packets: city index, selected building metrics, representative labels, representative papers, and bridge/street evidence. Raw full graph data is not sent.
- The city navigator can answer a research question and return focused building IDs, bridge IDs, route steps, confidence, and model availability.

Quality And Verification

- Backend tests cover OpenAlex normalization, deduplication, edge formulas, building/floor generation, bridge thresholds, street generation, seeded-city generation, and Groq JSON parsing.
- Frontend tests cover API loading, visual mappings, layer toggles, inspectors, labels, selection highlighting, seeded-city controls, and navigation behavior.
- Latest verification before this report: 40 backend tests passed, 32 frontend tests passed, and the frontend production build passed.

Key Design Decisions

- The main 3D scene stays at building level so the city remains readable at 1000 papers. Paper nodes appear only in focused interior or inspector views.
- Color represents semantic meaning. Activation is shown with glow and shade, not by changing domain identity.
- Bridges and streets are separated because they answer different questions: bridges show strong research relationships, while streets preserve weaker contextual routes.
- Inspectors expose original labels and paper titles so summaries and visual links are traceable back to data.

What Could Be Done Next

Data And Graph Quality

- Add embedding-based similarity for titles and abstracts to improve semantic edges beyond token overlap.
- Separate citation direction from similarity edges so influence and topical relatedness can be viewed independently.
- Improve seeded expansion with user-controlled breadth, depth, year range, and relatedness thresholds.
- Cache OpenAlex responses and store build provenance so every city can be reproduced.

Community And Layout

- Add hierarchical communities: districts, neighborhoods, buildings, floors, and paper nodes.
- Add a toggle between semantic-domain colors and modularity/community colors.
- Implement a more faithful Graph City street-layout pass using research relevance as the road weight, then compare it against the current bridges-and-streets representation.
- Improve placement so related buildings cluster into district-like groups and labels collide less.

User Workflows

- Add a small pre-city seed graph preview for the initial 10 papers before expanding to a 1000-paper city.
- Add paper search, filters by year/topic/venue, and one-click jump from search results to the corresponding building.
- Add saved tours and exportable evidence reports for presentations.

LLM And Evaluation

- Add retrieval over paper abstracts for deeper answers inside a building or across a bridge.
- Calibrate LLM confidence against evidence coverage and require citations from selected paper titles.
- Add evaluator tests for navigation quality on known seed sets and known research

questions.

Production Readiness

- Run seeded-city builds as background jobs rather than blocking an API request.
- Persist multiple user-generated cities with shareable IDs.
- Add deployment configuration, monitoring, retry logic, and rate-limit handling.
- Split frontend bundles so Three.js-heavy views load only when needed.

Conclusion

Research Graph City now demonstrates the core concept: a research-paper graph can become a navigable 3D city with meaningful buildings, floors, bridges, streets, colors, labels, and LLM-guided inspection. The next phase should focus on stronger graph semantics, better spatial layout, deeper retrieval, and workflows for creating and comparing multiple seeded cities.